Advanced Scenario Modeling in Excel

(Build structured scenario and sensitivity models)

Overview

This course uses an artifact sequence approach in which participants transform a structured dataset into a decision-support model with multiple scenarios and sensitivity outputs. Learning progresses from defining assumptions and drivers, to building scenario logic and comparison structures, and finally generating visual outputs for decision review. Each topic builds on the same model artifact so participants understand how assumption changes affect results. This mirrors real planning workflows and enables participants to reproduce reliable scenario models for operational and strategic decisions.

Objectives

Upon completion of the course, participants will be able to design structured scenario models in Excel that support comparative decision-making. This outcome emphasizes managing assumptions, controlling calculation logic, and interpreting sensitivity outputs with clarity and consistency.

Specifically, participants will be able to:

- design assumption-driven models with multiple scenarios
- implement structured comparison logic across scenarios
- construct sensitivity analysis outputs for decision review
- evaluate model reliability under changing inputs

Who Should Participate

- Managers involved in planning and forecasting decisions.
- Senior analysts building comparative models.
- Operations and strategy team members.
- Professionals responsible for scenario-based reporting.

Key Topics

I. Defining Assumptions and Drivers
   a. Identifying key input variables influencing outcomes
   b. Separating controllable drivers from calculated metrics
   c. Structuring centralized assumption inputs
   d. Designing transparent assumption tracking

II. Building Scenario Logic
   a. Designing parallel scenario calculation structures
   b. Implementing controlled conditional logic
   c. Managing comparison outputs across scenarios
   d. Preventing logic duplication errors

III. Implementing Sensitivity Analysis
   a. Designing structured input variation ranges
   b. Measuring output response to input shifts
   c. Interpreting impact gradients
   d. Ensuring consistent recalculation behavior

IV. Visualizing Scenario Outputs
   a. Selecting appropriate chart types for comparison
   b. Maintaining visual hierarchy and scaling consistency
   c. Avoiding misleading graphical emphasis
   d. Preparing visuals for executive interpretation

V. Validating and Stress Testing Models
   a. Auditing outputs against base calculations
   b. Testing extreme assumption values
   c. Identifying instability in formulas
   d. Preparing models for recurring planning cycle